Elementary Analysis

Chapter 2, Section 16

Decimal Expansions of Real Numbers
Exercises 16.1–16.10, with every subpart

Mathjock

Text and numbering. Kenneth A. Ross, *Elementary Analysis: The Theory of Calculus*, second edition (Springer, 2013), Chapter 2. Exercise numbers follow the copy already linked on Mathjock. These are independently written worked solutions, not the publisher's solutions manual. Each exercise is identified by a short mathematical description.

Reading philosophy. Every argument states its goal, records the assumptions being used, and explains the intermediate algebra or logical inference. Definitions, sign conditions, and both directions of an "if and only if" are made explicit. A cited rule is named at the point where it is used.

Conventions. $\mathbb{N} = \{1, 2, 3, \ldots\}$, $\mathbb{Z} = \{\ldots, -2, -1, 0, 1, 2, \ldots\}$, and $\mathbb{Q} = \{p/q : p, q \in \mathbb{Z},\ q \neq 0\}$. Unless a different domain is stated, sequence indices run through $\mathbb{N}$ and all terms and finite limits are real. The empty sum is 0, and $0! = 1$.

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16 Decimal expansions of real numbers

A bar marks exactly the digits that repeat. Thus $0.0\overline{2} = 0.02222\ldots$, whereas $0.\overline{02} = 0.020202\ldots$. A terminating decimal is understood to continue with zeros. For a decimal $K.d_1d_2\cdots$, its value means the convergent series

$$K + \sum_{j=1}^{\infty} d_j 10^{-j}, \qquad 0 \le d_j \le 9.$$

Convergence follows by comparison with the geometric series $\sum_{j=1}^{\infty} 9 \cdot 10^{-j} = 1$. Theorem 16.3 describes the only possible ambiguity: a positive terminating decimal also has a representation ending in nines.

16.1 Exercise 16.1: Two expansions for eleven fourths

What we must show. Verify both decimal representations in part (a), and identify which one long division produces in part (b).

(a) Evaluate the expansion ending in zeros.

The expression $2.75\overline{0}$ means

$$2 + \frac{7}{10} + \frac{5}{100} = \frac{200 + 70 + 5}{100} = \frac{275}{100} = \frac{11}{4}.$$

Every remaining term is zero, so there is no additional contribution.

(a) Evaluate the expansion ending in nines.

In $2.74\overline{9}$ the first nine is in the third decimal place. The sum of the nines is

$$\sum_{j=3}^{\infty} 9 \cdot 10^{-j} = 9 \cdot 10^{-3} \sum_{k=0}^{\infty} (10^{-1})^k = \frac{9 \cdot 10^{-3}}{1 - 10^{-1}} = \frac{1}{100}.$$

Consequently

$$2.74\overline{9} = 2 + \frac{7}{10} + \frac{4}{100} + \frac{1}{100} = 2 + \frac{7}{10} + \frac{5}{100} = \frac{11}{4}.$$

The extra tail of nines supplies exactly the missing hundredth.

(b) Carry out the long division.

First $11 = 4 \cdot 2 + 3$, so the integer part is 2 and the initial remainder is 3. Multiplying the remainder by ten gives

$$30 = 4 \cdot 7 + 2, \qquad 20 = 4 \cdot 5 + 0.$$

The successive decimal digits are therefore 7 and 5, after which the remainder is zero. All subsequent divisions have $10 \cdot 0 = 4 \cdot 0 + 0$, so all subsequent digits are zero.

Conclusion. Both $2.74\overline{9}$ and $2.75\overline{0}$ equal $11/4$. The long division algorithm produces $2.75\overline{0}$.

16.2 Exercise 16.2: Trading a terminating tail for a tail of nines

What we must show. Verify the two identities in the opening paragraph of the proof of Theorem 16.3: the identity for a positive integer, and the identity for a decimal with a last nonzero digit.

First calculate a general tail of nines.

For any integer $r \geq 0$, a geometric series gives

$$\sum_{j=r+1}^{\infty} 9 \cdot 10^{-j} = \frac{9 \cdot 10^{-(r+1)}}{1 - 10^{-1}} = 10^{-r}.$$

In particular, when $r = 0$ this tail has value 1.

A positive integer has the asserted alternative representation.

Let K be a positive integer. Then $K - 1$ is a nonnegative integer, so it is an allowed integer part for a nonnegative decimal. The tail formula with $r = 0$ gives

$$(K-1).\overline{9} = (K-1) + \sum_{j=1}^{\infty} 9 \cdot 10^{-j} = (K-1) + 1 = K = K.\overline{0}.$$

These digit strings are distinct even though their values agree.

A last nonzero digit can be reduced by one.

Suppose

$$x = K.d_1 \cdots d_{r-1} d_r \overline{0}, \qquad r \geq 1, \qquad d_r \in \{1, \ldots, 9\}.$$

The condition $d_r \neq 0$ makes $d_r - 1$ an allowed digit. The value of the proposed other expansion is

$$\begin{aligned} K + \sum_{j=1}^{r-1} d_j 10^{-j} + (d_r - 1)10^{-r} + \sum_{j=r+1}^{\infty} 9 \cdot 10^{-j} \\ = K + \sum_{j=1}^{r-1} d_j 10^{-j} + (d_r - 1)10^{-r} + 10^{-r} \\ = K + \sum_{j=1}^{r-1} d_j 10^{-j} + d_r 10^{-r} = x. \end{aligned}$$

Thus decreasing the last nonzero digit by one and replacing the following zeros by nines preserves the represented number.

Check the zero case separately.

The first identity requires $K > 0$, and the second requires a last nonzero digit. Neither condition holds for $0.\overline{0}$. Indeed, if a nonnegative decimal represents zero, its nonnegative integer part must be zero. Every digit must then be zero too: a positive digit d_j would contribute the positive amount $d_j 10^{-j}$ to a sum of nonnegative terms. Hence zero has just the expansion $0.\overline{0}$.

Conclusion. The two claimed alternative expansions follow from the geometric tail identity. Every positive terminating decimal has the indicated second expansion; zero is the exceptional case.

16.3 Exercise 16.3: Strict comparison of convergent nonnegative series

What we must show. Suppose $\sum a_n$ and $\sum b_n$ converge, all their terms are nonnegative, $a_n \le b_n$ for every n, and $a_m < b_m$ at at least one index m. Prove the inequality of their sums is strict.

Plan. Keep the positive gap at that one index when passing from finite sums to infinite sums. A strict inequality at each partial sum alone would not guarantee a strict inequality of limits.

Name the sums and the fixed positive gap.

Write

$$A = \sum_{n=1}^{\infty} a_n, \qquad B = \sum_{n=1}^{\infty} b_n, \qquad \delta = b_m - a_m > 0.$$

Both A and B are finite by hypothesis. For every $N \ge m$,

$$\sum_{n=1}^{N} b_n - \sum_{n=1}^{N} a_n = \sum_{n=1}^{N} (b_n - a_n) \ge b_m - a_m = \delta.$$

The inequality holds because all the other differences are nonnegative; deleting them cannot increase the sum.

Take the limit with the gap still present.

The difference rule for limits gives

$$B - A = \lim_{N\to\infty} \left(\sum_{n=1}^{N} b_n - \sum_{n=1}^{N} a_n \right) \ge \delta > 0.$$

Here the weak inequality is preserved by passage to finite limits, and the final strict inequality comes from the fixed number δ, which does not shrink with N.

Conclusion. $\displaystyle\sum_{n=1}^{\infty} a_n < \sum_{n=1}^{\infty} b_n.$

16.4 Exercise 16.4: Converting six repeating decimals to fractions

What we must show. Express each given repeating decimal as a fraction of integers.

Plan. Group the repeated block as an integer and use a geometric series. A block of r digits with integer value D, repeated immediately after the decimal point, represents

$$\frac{D}{10^r} + \frac{D}{10^{2r}} + \cdots = \frac{D10^{-r}}{1 - 10^{-r}} = \frac{D}{10^r - 1}.$$

A leading zero counts as a digit of the block: the block 02 has length two and integer value 2.

(a) The decimal $0.\overline{2}$.

Here the block has length one and value 2, so

$$0.\overline{2} = \frac{2}{10 - 1} = \boxed{\frac{2}{9}}.$$

(b) The decimal $0.0\overline{2}$.

Only the digit 2 repeats; the first decimal digit is the single nonrepeating zero. Moving the decimal point one place right gives $10x = 0.\overline{2} = 2/9$, where $x = 0.0\overline{2}$. Therefore

$$x = \frac{1}{10}\frac{2}{9} = \frac{2}{90} = \boxed{\frac{1}{45}}.$$

This differs from part (c), where the zero repeats as part of the block.

(c) The decimal $0.\overline{02}$.

The repeated block has length two, so

$$0.\overline{02} = \frac{2}{10^2 - 1} = \boxed{\frac{2}{99}}.$$

(d) The decimal $3.\overline{14}$.

Separate the integer part before evaluating the length-two block:

$$3.\overline{14} = 3 + \frac{14}{99} = \frac{297 + 14}{99} = \boxed{\frac{311}{99}}.$$

(e) The decimal $0.\overline{10}$.

The block is 10, of length two and integer value 10. Hence

$$0.\overline{10} = \boxed{\frac{10}{99}}.$$

(f) The decimal $0.\overline{1492}$.

This block has length four and integer value 1492, giving

$$0.\overline{1492} = \frac{1492}{10^4 - 1} = \boxed{\frac{1492}{9999}}.$$

Conclusion. The six fractions, in order, are 2/9, 1/45, 2/99, 311/99, 10/99, and 1492/9999.

16.5 Exercise 16.5: Decimal expansions of six rational numbers

What we must show. Find the decimal expansions of the six prescribed fractions. For terminating values, include both expansions supplied by Theorem 16.3.

(a) Expand $1/8$.

At each stage divide ten times the current remainder by 8:

$$10 = 8 \cdot 1 + 2, \qquad 20 = 8 \cdot 2 + 4, \qquad 40 = 8 \cdot 5 + 0.$$

The quotients give digits $1, 2, 5$. A zero remainder produces zeros forever. Reducing the last nonzero digit by one gives the other expansion, so

$$\boxed{\frac{1}{8} = 0.125\overline{0} = 0.124\overline{9}}.$$

(b) Expand $1/16$**.**

The successive divisions are

$$10 = 16 \cdot 0 + 10, \quad 100 = 16 \cdot 6 + 4,$$
$$40 = 16 \cdot 2 + 8, \qquad 80 = 16 \cdot 5 + 0.$$

The initial quotient zero matters: it is the first decimal digit. The digits are $0, 6, 2, 5$, followed by zeros. Thus

$$\boxed{\frac{1}{16} = 0.0625\overline{0} = 0.0624\overline{9}}.$$

(c) Expand $2/3$**.**

The first division is $20 = 3 \cdot 6 + 2$. The new remainder is the original remainder 2, so the same digit 6 and remainder 2 recur at every step. Therefore

$$\boxed{\frac{2}{3} = 0.\overline{6}}.$$

As a check, the geometric-series formula gives $0.\overline{6} = 6/9 = 2/3$.

(d) Expand $7/9$**.**

Here $70 = 9 \cdot 7 + 7$. The remainder 7 returns immediately, so every digit is 7:

$$\boxed{\frac{7}{9} = 0.\overline{7}}.$$

(e) Expand $6/11$**.**

The divisions $60 = 11 \cdot 5 + 5$ and $50 = 11 \cdot 4 + 6$ return to the initial remainder 6 after the two digits $5, 4$. Hence

$$\boxed{\frac{6}{11} = 0.\overline{54}}.$$

The block formula confirms $54/99 = 6/11$.

(f) Expand $22/7$**.**

First $22 = 7 \cdot 3 + 1$, giving integer part 3 and remainder 1. The next six divisions are

$$10 = 7 \cdot 1 + 3, \quad 30 = 7 \cdot 4 + 2,$$
$$20 = 7 \cdot 2 + 6, \quad 60 = 7 \cdot 8 + 4,$$
$$40 = 7 \cdot 5 + 5, \quad 50 = 7 \cdot 7 + 1.$$

The remainder has returned to 1, so the six-digit block repeats:

$$\boxed{\frac{22}{7} = 3.\overline{142857}}.$$

Account for all possible expansions.

The last four displayed expansions have neither an eventual tail of zeros nor an eventual tail of nines. Theorem 16.3 says they are unique. The first two are positive terminating decimals, so the two representations displayed in each of (a) and (b) are their only representations.

Conclusion. The expansions are the six boxed answers above, including both representations in the two terminating cases.

16.6 Exercise 16.6: The cyclic decimal blocks for the sevenths

What we must show. Find the expansions of $1/7, 2/7, \ldots, 6/7$ and explain the pattern in their repeating blocks.

Record the long division rule once for each possible nonzero remainder.

For a current remainder r, the next digit d and next remainder r' satisfy $10r = 7d + r'$ with $0 \le r' < 7$. The six possible nonzero remainders give

r	1	2	3	4	5	6
d	1	2	4	5	7	8
r'	3	6	2	5	1	4

For instance, the entry $r = 3$ comes from $30 = 7 \cdot 4 + 2$, so the next digit is 4 and the next remainder is 2.

Follow the cycle starting from remainder one.

The remainder rule traces

$$1 \longmapsto 3 \longmapsto 2 \longmapsto 6 \longmapsto 4 \longmapsto 5 \longmapsto 1.$$

The digits produced on these six steps are respectively $1, 4, 2, 8, 5, 7$. All six remainders in the cycle are distinct, so returning to the initial remainder takes exactly six steps.

Start the same cycle at each of the other remainders.

The fraction $a/7$ begins with remainder a, for $a = 1, \ldots, 6$. Starting the displayed cycle at that remainder gives

Fraction	Remainders before the successive digits	Expansion
$1/7$	$1, 3, 2, 6, 4, 5$	$0.\overline{142857}$
$2/7$	$2, 6, 4, 5, 1, 3$	$0.\overline{285714}$
$3/7$	$3, 2, 6, 4, 5, 1$	$0.\overline{428571}$
$4/7$	$4, 5, 1, 3, 2, 6$	$0.\overline{571428}$
$5/7$	$5, 1, 3, 2, 6, 4$	$0.\overline{714285}$
$6/7$	$6, 4, 5, 1, 3, 2$	$0.\overline{857142}$

Each row returns to its initial remainder after the six digits, so the same block repeats forever. These expansions are unique because none has an eventual tail of zeros or nines.

Conclusion. The six blocks are cyclic rotations of 142857. Long division uses the same cycle of six nonzero remainders, with a different starting position for each fraction.

16.7 Exercise 16.7: The decimal formed by concatenating integers

What we must show. Decide whether the number

$$c = 0.12345678910111213141516171819202122223\cdots$$

is rational. Here its digits are obtained by writing $1, 2, 3, \ldots$ in their ordinary decimal notation and concatenating those strings.

Plan. Prove this digit string is not eventually periodic. It has arbitrarily long strings of zeros arbitrarily far out, but it also has nonzero digits arbitrarily far out. Both facts are needed: a decimal ending in zeros would have the first property alone.

Find long zero strings after any prescribed position.

For every positive integer m, the integer 10^m appears in the concatenation. Its ordinary decimal notation is the digit 1 followed by m zeros. As m increases, the starting positions of these integers go arbitrarily far out: before writing 10^m, we have already written the $10^m - 1$ preceding positive integers, each of which contributes at least one digit. Thus, after any specified digit position and for any specified length p, there is a string of at least p consecutive zeros. Choose $m \geq p$ large enough that the occurrence of 10^m starts after that position.

Find nonzero digits after any prescribed position.

The initial digit 1 in each of those occurrences of 10^m is nonzero, and these occurrences start arbitrarily far out. Consequently the decimal does not eventually consist only of zeros. It does not eventually consist only of nines either, since zero strings also occur arbitrarily far out.

Rule out an eventual repeating block.

Suppose the digits were periodic with period $p \geq 1$ after some position N. If the repeating block contained only zeros, all digits beyond N would be zero, contradicting the preceding step. Otherwise the block contains a nonzero digit. Every run of p consecutive positions beyond N passes through every position of the repeating block once, so it must contain a nonzero digit. But the first step supplies a run of p zeros entirely beyond N. This is a contradiction in both cases.

Apply the decimal characterization of rational numbers.

The given decimal is not eventually periodic. By Theorem 16.5, a rational number has a repeating decimal expansion; by Theorem 16.3 its only possible second expansion would have tails of zeros and nines. Those tails have already been excluded here. Therefore the displayed decimal cannot represent a rational number.

Conclusion. The concatenation decimal is irrational.

16.8 Exercise 16.8: Constructing a number outside a proposed list

What we must show. Given any sequence (s_n) in $(0,1)$, with selected expansions $s_n = 0.d_1^{(n)}d_2^{(n)}\cdots$, define

$$e_n = \begin{cases} 6, & d_n^{(n)} \neq 6, \\ 7, & d_n^{(n)} = 6. \end{cases}$$

Show that $0.e_1e_2\cdots$ represents a number $y \in (0,1)$ different from every s_n.

Check that the new digit string represents a number.

Every e_n is either 6 or 7, hence an allowed decimal digit. Define the partial sums $y_N = \sum_{n=1}^{N} e_n 10^{-n}$. They increase and are bounded above by the convergent geometric series $\sum_{n=1}^{\infty} 7 \cdot 10^{-n} = 7/9$. The monotone sequence theorem gives a finite limit y, which by the definition of a decimal is the number $0.e_1e_2\cdots$. Termwise comparison of the finite partial sums and passage to limits give

$$\frac{2}{3} = \sum_{n=1}^{\infty} 6 \cdot 10^{-n} \leq y \leq \sum_{n=1}^{\infty} 7 \cdot 10^{-n} = \frac{7}{9}.$$

Because $0 < 2/3 \leq 7/9 < 1$, these bounds imply $y \in (0,1)$.

Verify the diagonal difference at every index.

Fix n. If $d_n^{(n)} \neq 6$, the construction gives $e_n = 6$, so $e_n \neq d_n^{(n)}$. If $d_n^{(n)} = 6$, it gives $e_n = 7$, so again $e_n \neq d_n^{(n)}$. Thus the new expansion differs from the selected expansion of s_n at least at its nth decimal digit.

Explain why different strings cannot secretly represent the same number.

Different decimal strings can represent the same real number in the zero-tail/nine-tail case of Theorem 16.3. That issue must be checked rather than ignored. Every digit of the expansion of y is 6 or 7, so this expansion ends neither in zeros nor in nines. Theorem 16.3 therefore makes it the unique expansion of y. If $y = s_n$, the selected expansion of s_n would be a second expansion of y differing at digit n, which is impossible. Hence $y \neq s_n$ for every n.

Interpret the conclusion for a supposed enumeration.

If a sequence listed every number in $(0,1)$, the number y constructed from that sequence would belong to $(0,1)$ and yet would be absent from the list. This contradiction proves that no such listing exists.

Conclusion. The constructed y lies in $(0,1)$ and differs from every listed term. Thus $(0,1)$ is uncountable.

16.9 Exercise 16.9: Existence of Euler's constant

What we must show. For

$$H_n = \sum_{k=1}^{n} \frac{1}{k}, \qquad \gamma_n = H_n - \log n = H_n - \int_1^n \frac{dt}{t},$$

prove (a) (γ_n) decreases, (b) $0 < \gamma_n \leq 1$, and (c) its limit exists and is finite. The logarithm here is natural.

(a) Subtract two neighboring terms without discarding the logarithm.

Since $H_{n+1} = H_n + 1/(n+1)$, we have

$$\begin{aligned}\gamma_n - \gamma_{n+1} &= H_n - \log n - H_n - \frac{1}{n+1} + \log(n+1) \\ &= \int_n^{n+1} \frac{dt}{t} - \frac{1}{n+1} \\ &= \int_n^{n+1} \left(\frac{1}{t} - \frac{1}{n+1}\right) dt.\end{aligned}$$

For $n \le t < n+1$ the integrand is positive, because $t < n+1$ implies $1/t > 1/(n+1)$. It is continuous and positive on the interior of this interval, so its integral is strictly positive. For an explicit lower estimate, on $[n, n+1/2]$ the integrand is at least $1/(n+1/2) - 1/(n+1) > 0$, and that interval has positive length $1/2$. Hence $\gamma_n > \gamma_{n+1}$ for every n.

(b) Bound the integral between two rectangle sums.

For $t \in [k, k+1]$ with $k \ge 1$, reciprocation gives

$$\frac{1}{k+1} \le \frac{1}{t} \le \frac{1}{k}.$$

Integrating over an interval of length one preserves these bounds:

$$\frac{1}{k+1} \le \int_k^{k+1} \frac{dt}{t} \le \frac{1}{k}.$$

For $n \ge 2$, sum from $k = 1$ to $k = n-1$ to obtain

$$H_n - 1 \le \int_1^n \frac{dt}{t} \le H_n - \frac{1}{n}.$$

Subtract the middle expression from H_n. Reversal of the inequalities under subtraction gives

$$\frac{1}{n} \le \gamma_n \le 1.$$

The same conclusion holds for $n = 1$ directly: $\gamma_1 = 1 - \log 1 = 1$. In particular $\gamma_n > 0$ at every index.

(c) Apply monotone convergence with a real lower bound.

Part (a) makes (γ_n) decreasing, and part (b) bounds it below by zero. The monotone sequence theorem therefore gives a finite real limit

$$\gamma = \lim_{n\to\infty} \gamma_n \in [0, 1].$$

The fact that the two pieces H_n and $\log n$ both grow without bound does not obstruct convergence of their difference. The monotonicity and bounds control that difference directly.

Conclusion. The sequence is strictly decreasing, satisfies $0 < \gamma_n \le 1$, and has a finite limit called Euler's constant.

16.10 Exercise 16.10: Irrationality of the indicated powers and roots of pi

What we must show. Starting with the irrationality of π^2 established in Example 6(b), determine what follows for π, $\sqrt{\pi}$, and $\sqrt[3]{\pi}$. Also address π^4 and distinguish the additional fact needed for that last conclusion.

Recall the elementary closure property of rational powers.

If $q = a/b$ is rational, with $a, b \in \mathbb{Z}$ and $b \neq 0$, then for every positive integer m,

$$q^m = \frac{a^m}{b^m} \in \mathbb{Q},$$

since the numerator and denominator are integers and $b^m \neq 0$. Consequently, if a positive integer power of a number is irrational, the number itself must be irrational.

Apply this property to π and to its square root.

If π were rational, its square would be rational, contrary to the given result. Therefore π is irrational. If $\sqrt{\pi}$ were rational, its square would be π and would be rational, giving the same contradiction. Equivalently, its fourth power is π^2, which is already known to be irrational. Thus $\sqrt{\pi}$ is irrational too.

Apply the same argument to the cube root.

If $\sqrt[3]{\pi}$ were rational, its cube π would be rational. Since π is irrational, this is impossible. One can also apply the closure property directly to the sixth power: $(\sqrt[3]{\pi})^6 = \pi^2$. Hence $\sqrt[3]{\pi}$ is irrational.

Explain what the premise alone says about π^4.

The implication just used runs from irrationality of a power to irrationality of its base. It cannot be reversed: the square of an irrational number can be rational. For a concrete comparison, let $x = \sqrt[4]{2}$. Then $x^2 = \sqrt{2}$ is irrational, but $x^4 = 2$ is rational. Thus the statement "x^2 is irrational" does not by itself imply "x^4 is irrational." In particular, the given irrationality of π^2 alone does not settle the question about π^4 by this rational-power argument.

Determine the actual status of π^4 using the stronger fact stated in the text.

Example 6(c) additionally states the deeper theorem that π is not algebraic. This means it is not a zero of any nonzero polynomial with integer coefficients. That stated theorem does settle the question: if $\pi^4 = a/b$ were rational, where $a, b \in \mathbb{Z}$ and $b \neq 0$, then

$$b\pi^4 - a = 0.$$

The polynomial $P(X) = bX^4 - a$ has integer coefficients and is nonzero because its leading coefficient b is nonzero. The displayed equation would make π algebraic, contradicting Example 6(c). Therefore π^4 is irrational.

This final step cites the stronger theorem expressly stated in the chapter; it is not a consequence of the irrationality of π^2 alone.

Conclusion. π, $\sqrt{\pi}$, and $\sqrt[3]{\pi}$ are irrational by elementary rational-power arguments from π^2. The number π^4 is also irrational, using the stronger nonalgebraicity result stated in Example 6(c).